

LAB 3 DELIVERABLES - COMPONENT MODELLING AND ARCHITECTURE

PATTERN SELECTION

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PROJECT TITLE : MULTI-CLOUD ASSET COST OPTIMIZATION PORTAL

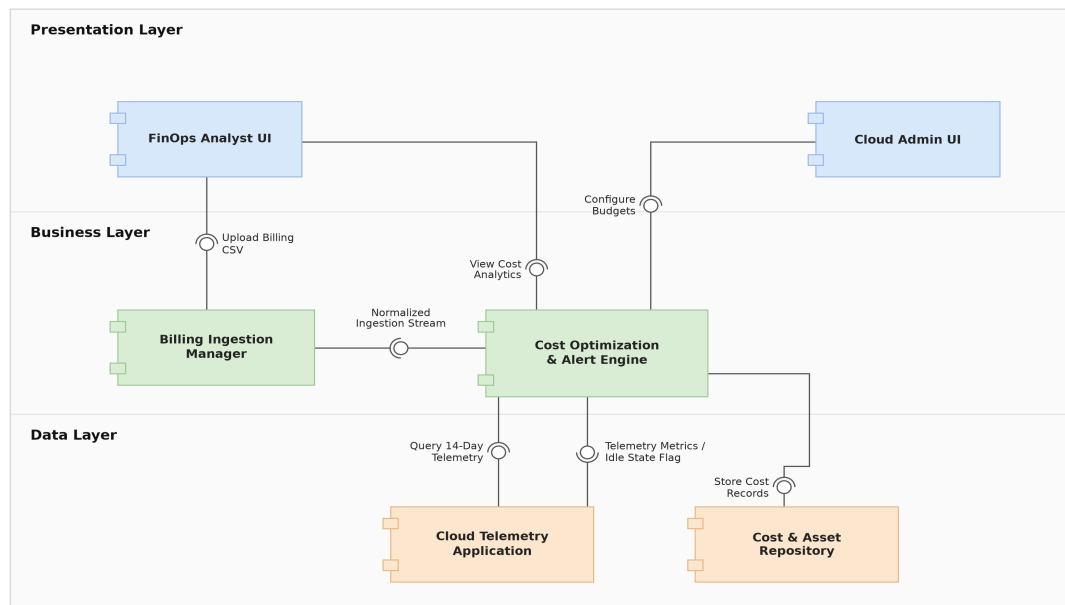
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DELIVERABLES

1. ARCHITECTURE SELECTION AND DIAGRAM
2. JUSTIFICATION

ARCHITECTURE SELECTION AND DIAGRAM



JUSTIFICATION

Architecture Selection: "We chose Layered Architecture for the Multi-Cloud Asset Cost Optimization Portal System."

Architectural Choice:

We selected the Layered Architecture pattern.

Two Reasons:

1. Clear Separation of Concerns: This pattern effectively isolates the user-facing interfaces (FinOps Analyst UI and Cloud Admin UI) from the complex backend telemetry ingestion, optimization heuristics, and alerting logic.
2. Supports Traditional Workflows: The step-by-step pipeline from multi-cloud billing CSV ingestion (Presentation), to stream normalization, idle heuristic scoring, and alert routing (Business), to persistent storage (Data) maps perfectly to horizontal layers.

Security Advantage:

The architecture provides a significant security advantage by preventing direct database access from the Presentation layer. Because tight coupling is managed strictly between adjacent layers, the FinOps Analyst UI and Cloud Admin UI cannot directly edit or query the Cost & Asset Repository, nor can they directly touch cloud provider IAM credentials. All cost analytical queries, budget thresholds, and alerting rules must pass through the Cost Optimization & Alert Engine in the Business layer, ensuring that every financial change and resource policy is securely validated, role-audited, and encrypted before hitting the Data layer.

Performance Benefit:

A key performance benefit of this architecture is that the Business layer can act as a buffer. The Billing Ingestion Manager can stream, batch-parse, and normalize multi-gigabyte billing files (AWS CUR, Azure Cost Details, GCP Billing) before passing normalized records to the Data layer's Cost & Asset Repository and the Cloud Telemetry Application. This prevents the heavy multi-cloud datastore and telemetry monitoring engines from experiencing I/O saturation or crashing under periodic surges of massive billing file uploads, ensuring the portal consistently satisfies its sub-2-minute dashboard update latency requirement (NFR-001).